

Experiment No.	01
Experiment Title	Working with Python packages – Numpy, Scipy, Scikit-Learn, Matplotlib
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1 Objective

- To explore the core functions and methods available in NumPy, Pandas, SciPy, Scikit-learn, and Matplotlib, covering array manipulation, data preprocessing, mathematical computing, machine learning workflows, and data visualization [CO1, K2].
- To download and explore public datasets (Loan Amount Prediction, Handwritten Character Recognition/MNIST, Email Spam Classification, Diabetes, and Iris) and identify the appropriate machine learning model category (Supervised, Unsupervised, Semi-supervised, Regression, or Classification) for each [CO1, K3].
- To identify the type of ML task associated with each dataset and to explore the steps of the machine learning workflow: loading the dataset, Exploratory Data Analysis and visualization, data preprocessing, feature selection, data splitting, and performance evaluation [CO1, K2].

2 Problem Statement

Explore the various functions and methods available in the Python libraries NumPy, Pandas, SciPy, Scikit-learn, and Matplotlib. Using these libraries, load and explore five public datasets – Iris, Loan Amount Prediction, Diabetes, Email Spam Classification, and MNIST – from repositories such as the UCI Machine Learning Repository and Kaggle. For each dataset, perform Exploratory Data Analysis (summary statistics, missing-value analysis, and visualizations such as histograms, correlation heatmaps, and class-distribution plots), and identify the type of machine learning task, a suitable feature-selection technique, and a suitable ML algorithm. This experiment is limited to library exploration, dataset analysis, and EDA; no model is trained.

3 Dataset Description

Loan Prediction Dataset

Dataset Name	Loan Prediction Dataset
Dataset Source	https://www.kaggle.com/datasets/altruistdelhite04/loan-prediction-problem-dataset
Number of Samples	614
Number of Features	13

Number of Classes	
Missing Values	149 (total, across all columns)
Train-Test Split	

Pima Indians Diabetes Dataset

Dataset Name	Diabetes Dataset
Dataset Source	https://archive.ics.uci.edu/ml/datasets/Pima+Indians+Diabetes
Number of Samples	768
Number of Features	9
Number of Classes	2 (Outcome: 0 = Non-Diabetic, 1 = Diabetic)
Missing Values	0
Train-Test Split	

Iris Dataset

Dataset Name	Iris Species Dataset
Dataset Source	https://www.kaggle.com/datasets/uciml/iris
Number of Samples	150
Number of Features	6 (Id, SepalLengthCm, SepalWidthCm, PetalLengthCm, PetalWidthCm, Species)
Number of Classes	3 (Iris-setosa, Iris-versicolor, Iris-virginica)
Missing Values	0
Train-Test Split	

Email Spam Classification Dataset

Dataset Name	SMS Spam Collection Dataset
Dataset Source	https://www.kaggle.com/datasets/uciml/sms-spam-collection-dataset
Number of Samples	5572
Number of Features	5 (v1 – label, v2 – message text, and 3 largely empty unnamed columns)
Number of Classes	2 (ham, spam)
Missing Values	16648 (total, mostly from the 3 unnamed columns)
Train-Test Split	

MNIST Dataset

Dataset Name	MNIST Handwritten Digit Dataset
Dataset Source	https://www.kaggle.com/datasets/oddrational/mnist-in-csv
Number of Samples	60000

Number of Features	785 (784 pixel columns + 1 label column)
Number of Classes	10
Missing Values	0
Train-Test Split	

4 Exploratory Data Analysis

EDA was performed in the notebook (`ex1.ipynb`) using two reusable routines: `eda_tabular` for tabular datasets and `eda_images` for the image dataset. These were run on the Loan Prediction, Diabetes, and MNIST datasets. The Iris and Email Spam datasets were used only for dataset description and ML task identification (Sections 3 and 5) as required by the experiment brief, and were not passed through the notebook’s EDA routines in this run; no figures are therefore available for them.

Loan Prediction Dataset

The dataset was loaded successfully and contained 614 samples across 13 columns. The initial inspection reported the following missing-value counts per column: Gender – 13, Married – 3, Dependents – 15, Self_Employed – 32, LoanAmount – 22, Loan_Amount_Term – 14, and Credit_History – 50, giving a total of 149 missing values, with 0 duplicate rows.

The subsequent step of the EDA routine, which separates numeric and categorical columns and generates the histogram and correlation-heatmap figures, raised a **`TypeError`**. The error originated from the `df.select_dtypes()` call and read: “string dtypes are not allowed, use ‘object’ instead”. As a result, **no histogram or correlation-heatmap figures were generated for the Loan Prediction dataset** in this run of the notebook. Only the shape, column-type, missing-value, and duplicate-row statistics reported above are available.

Diabetes Dataset

The diabetes dataset contains 768 samples and 9 numeric features, with no missing values and no duplicate rows reported by `isnull().sum()` and `duplicated().sum()`.

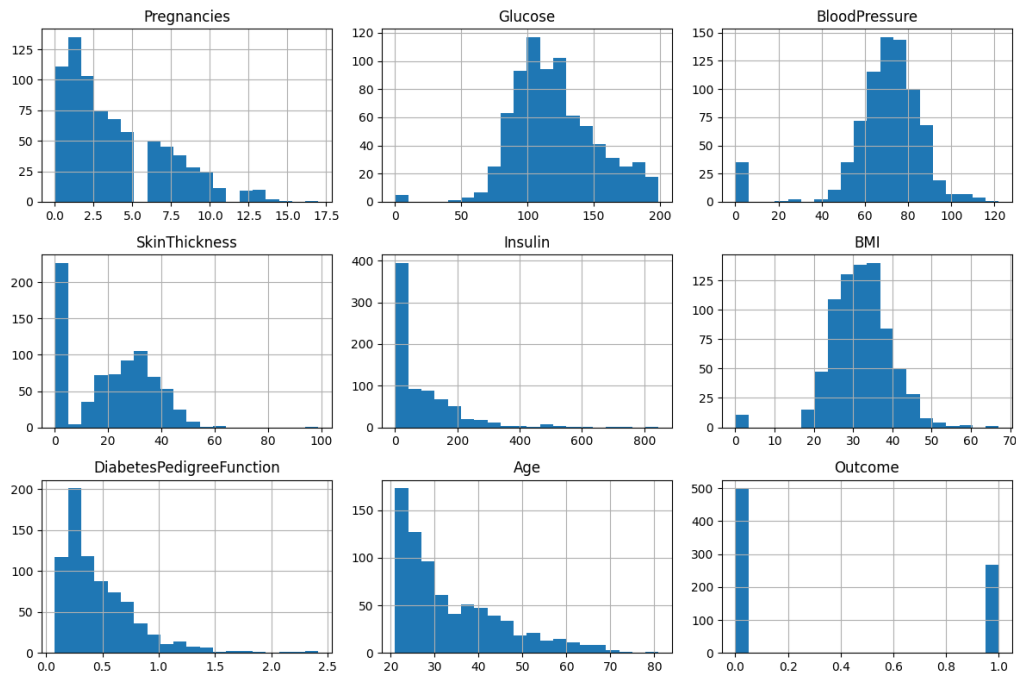


Figure 1: Histogram plots of the numeric features of the Diabetes dataset

Inference: The histograms show that most features (e.g. Glucose, BMI, Age) are right-skewed rather than normally distributed. Several features – notably SkinThickness and Insulin – show a large spike of zero values, which likely correspond to missing measurements that were recorded as 0 rather than as NaN. Pregnancies and Age are also concentrated at lower values with long right tails.

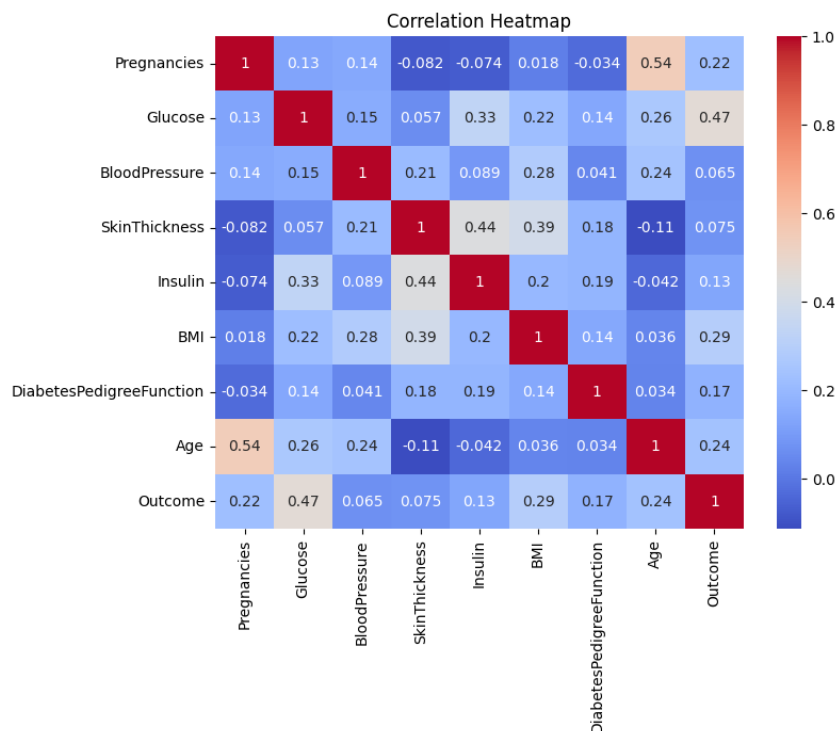


Figure 2: Correlation heatmap of the Diabetes dataset features

Inference: Glucose shows the strongest positive correlation with the Outcome variable, consistent with it being a key clinical indicator of diabetes. BMI and Age also show moderate positive correlation with Outcome. Most feature pairs show weak to moderate correlation with each other, indicating limited multicollinearity among the predictors.

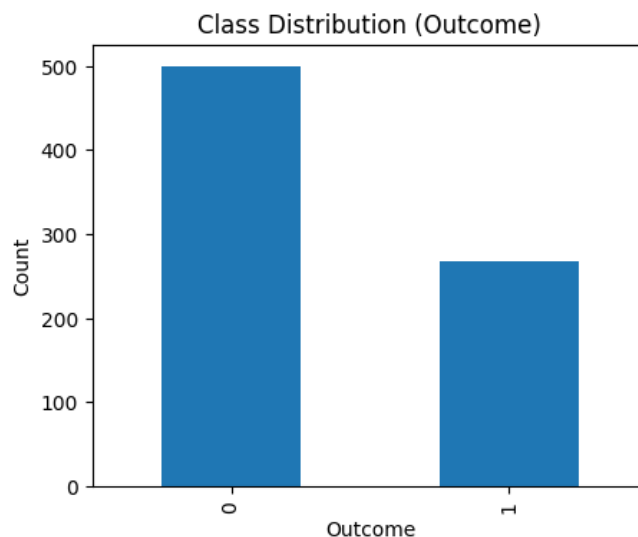


Figure 3: Class distribution of the Outcome variable in the Diabetes dataset

Inference: The class distribution is imbalanced: non-diabetic cases (Outcome = 0) considerably outnumber diabetic cases (Outcome = 1). This imbalance should be accounted for during model training and evaluation, for example by using stratified splitting or class-weighted metrics.

MNIST Dataset

The MNIST dataset contains 60,000 samples and 785 columns (1 label column and 784 pixel columns), with 0 missing values and 0 duplicate rows.

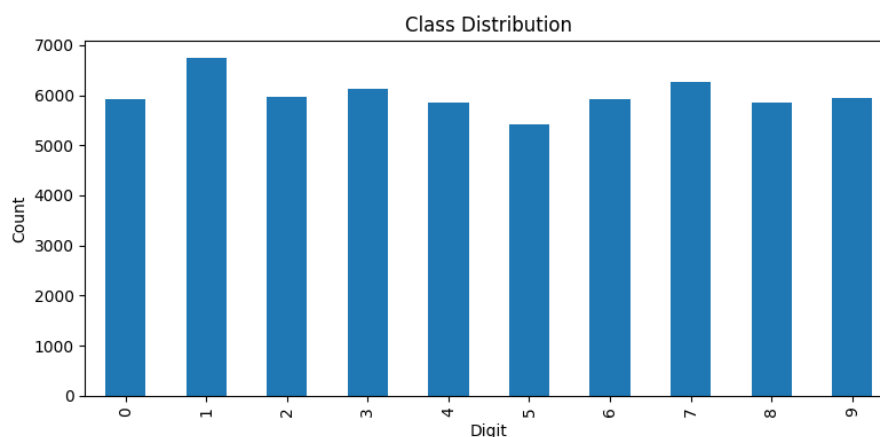


Figure 4: Class distribution of digit labels in the MNIST dataset

Inference: The ten digit classes (0–9) are fairly evenly distributed, each containing

between roughly 5,400 and 6,700 samples. This near-uniform distribution means accuracy is a reasonably fair evaluation metric for classifiers trained on this dataset.

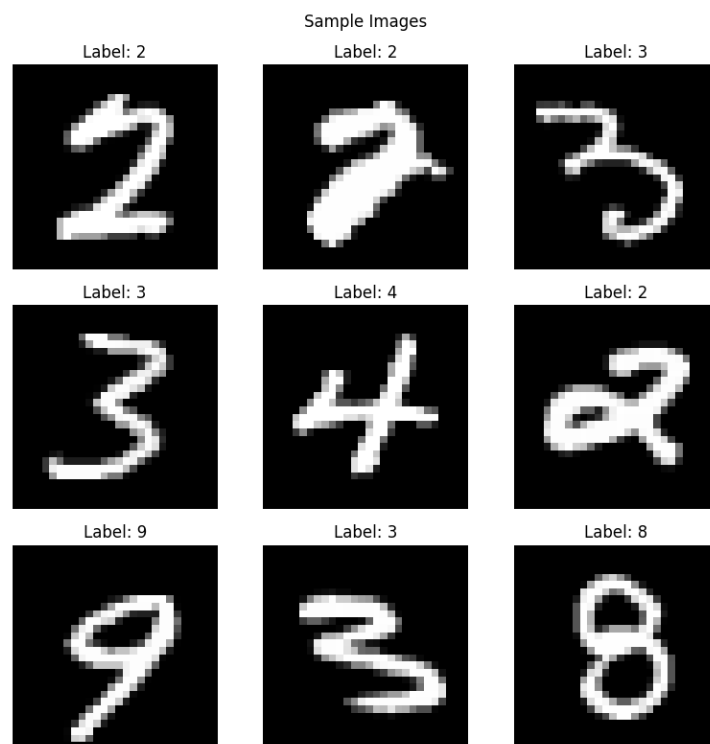


Figure 5: Randomly sampled images from the MNIST dataset

Inference: The sampled images confirm that each record is a 28×28 grayscale handwritten digit correctly labeled with its corresponding class. The digits show natural variation in stroke thickness and shape, which is representative of real handwriting.

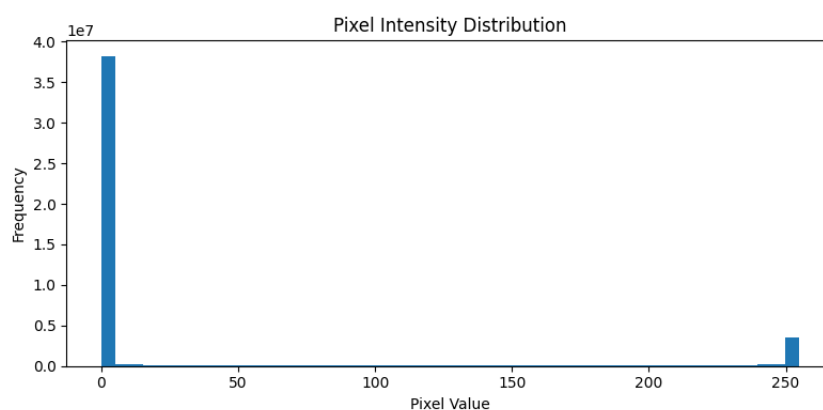


Figure 6: Pixel intensity distribution across the MNIST dataset

Inference: The pixel intensity distribution is heavily skewed toward 0, since most of each image is a black background. The Pixel Statistics computed from the notebook confirm this: minimum = 0, maximum = 255, mean = 33.32, and standard deviation = 78.57, indicating that only a small fraction of pixels (the digit strokes) carry high intensity values.

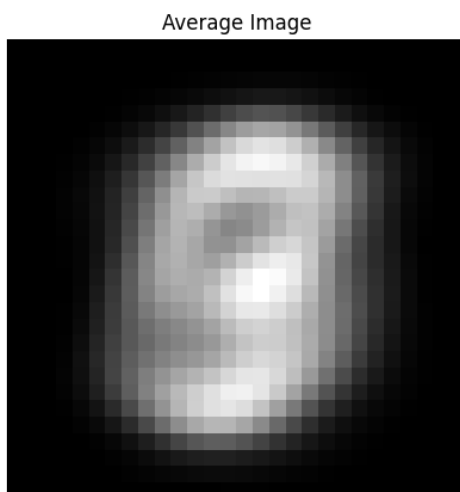


Figure 7: Average image computed across all MNIST samples

Inference: The average image shows a faint, blurred digit-like shape concentrated in the center of the frame, confirming that handwritten digits in this dataset are consistently centered and similarly scaled across samples.

5 ML Task Identification

As required by the experiment brief, the type of ML task, a suitable feature-selection technique, and a suitable ML algorithm are identified below for each of the five reference datasets.

Dataset	Type of ML Task	Feature Selection Technique	Suitable ML Algorithm
Iris Dataset	Classification	ANOVA / SelectKBest	Decision Tree, SVM, KNN
Loan Amount Prediction	Classification	Chi-Square / SelectKBest	Logistic Regression, Random Forest
Predicting Diabetes	Classification	ANOVA / SelectKBest	Logistic Regression, Naive Bayes, KNN
Classification of Email Spam	Classification	Chi-Square	Naive Bayes, SVM
Handwritten Character Recognition / MNIST	Classification	PCA (Optional)	KNN, CNN, Random Forest

6 Experimental Setup

Python Version	3.13.1
Scikit-Learn Version	
IDE	Jupyter Notebook
Operating System	Windows
Hardware	

7 Discussion

This experiment covered library exploration and dataset analysis using different types of datasets, in line with the experiment brief. Exploratory Data Analysis was performed on the Loan Prediction, Diabetes, and MNIST datasets to understand their structure, quality, and distribution. For the numerical datasets, summary statistics, histograms, and correlation heatmaps were used; for the image dataset, class distribution, sample images, pixel intensity distribution, and the average image were analyzed. The Iris and Email Spam datasets were additionally profiled (Section 3) and used for ML task identification (Section 5), though they were not run through the notebook’s EDA routines in this experiment.

A practical issue was encountered while analyzing the Loan Prediction dataset: the reusable `eda_tabular` function raised a `TypeError` at the `df.select_dtypes()` call due to a version-specific pandas restriction on string-type arguments, which prevented the histogram and correlation-heatmap figures from being generated for that dataset. This highlights the importance of validating helper functions against the specific library versions being used, and of not assuming reusable code will behave identically across datasets and environments.

All five reference datasets from the experiment brief (Iris, Loan Prediction, Diabetes, Email Spam, and MNIST) correspond to classification tasks. Suitable feature-selection techniques include ANOVA/SelectKBest for continuous numerical features (Iris, Diabetes), Chi-Square/SelectKBest for categorical or count-based features (Loan Prediction, Email Spam), and optional PCA for the high-dimensional MNIST image data. Candidate algorithms range from Decision Trees, SVM, and KNN for the smaller tabular datasets to CNN and Random Forest for MNIST.

8 Conclusion

The experiment provided hands-on exposure to the core Python libraries used in machine learning – NumPy, Pandas, SciPy, Scikit-learn, and Matplotlib. Exploratory Data Analysis was carried out on the Diabetes and MNIST datasets, producing histograms, correlation heatmaps, class-distribution plots, and image-based visualizations, while the Loan Prediction dataset was partially analyzed before a runtime error interrupted the pipeline. The Iris and Email Spam datasets were described and included in the ML task identification exercise. Suitable machine learning tasks and feature-selection techniques were identified for all five datasets, establishing a working foundation in dataset exploration for the subsequent laboratory experiments.

9 References

1. scikit-learn developers, "scikit-learn: Machine Learning in Python." [Online]. Available: <https://scikit-learn.org/stable/>. [Accessed: Aug. 2, 2026].
2. The pandas development team, "pandas documentation." [Online]. Available: <https://pandas.pydata.org/docs/>. [Accessed: Aug. 2, 2026].
3. The Matplotlib development team, "Matplotlib documentation." [Online]. Available: <https://matplotlib.org/stable/index.html>. [Accessed: Aug. 2, 2026].
4. M. Waskom, "seaborn: statistical data visualization." [Online]. Available: <https://seaborn.pydata.org/>. [Accessed: Aug. 2, 2026].
5. UCI Machine Learning Repository, "Pima Indians Diabetes Dataset." [Online]. Available: <https://archive.ics.uci.edu/ml/datasets/Pima+Indians+Diabetes>. [Accessed: Aug. 2, 2026].
6. Kaggle, "Loan Prediction Problem Dataset." [Online]. Available: <https://www.kaggle.com/datasets/altruistdelhite04/loan-prediction-problem-dataset>. [Accessed: Aug. 2, 2026].
7. Kaggle, "Iris Species." [Online]. Available: <https://www.kaggle.com/datasets/uciml/iris>. [Accessed: Aug. 2, 2026].
8. Kaggle, "SMS Spam Collection Dataset." [Online]. Available: <https://www.kaggle.com/datasets/uciml/sms-spam-collection-dataset>. [Accessed: Aug. 2, 2026].
9. Kaggle, "MNIST in CSV." [Online]. Available: <https://www.kaggle.com/datasets/oddrational/mnist-in-csv>. [Accessed: Aug. 2, 2026].